

ONTARIO · QISKIT FALL FEST

Can quantum kernels estimate **fire size**?

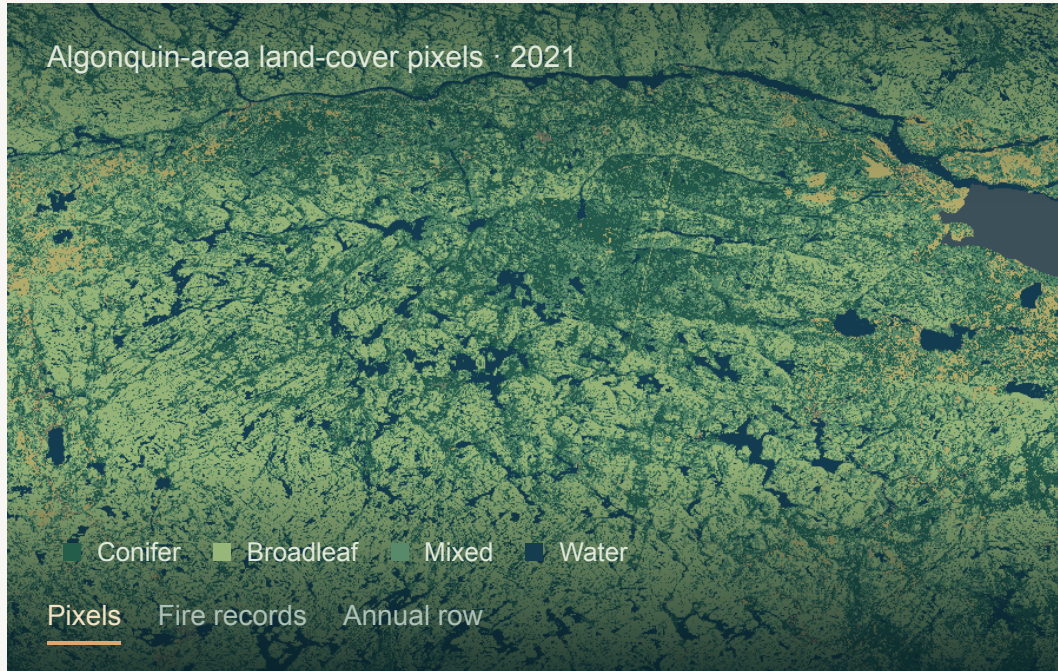
Annual mean size. Four qubits. A matched comparison.

657 ha / recorded fire
Ontario, 2021



Full Ontario boundary · grey = unmapped woodland · 2021 recorded locations

One Ontario year. One row.



NRCan · NFDB

784,447 ha ÷ 1,194 fires

657 ha/fire

ECCC · monthly climate



Ten annual / seasonal summaries · schematic

Join by year → one annual observation



31 training years · 1988–2018

6 reused years · 2019–2024

Woodland = context / separate ablation. Final predictors use climate.

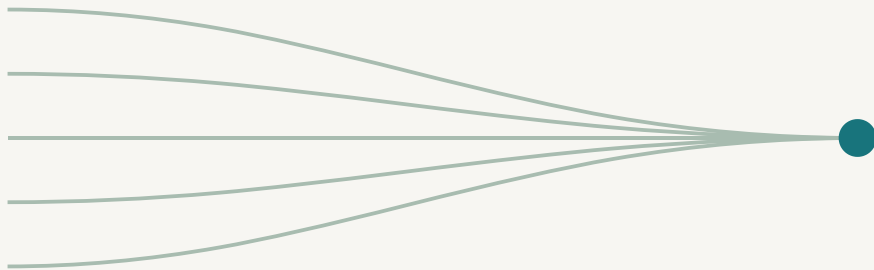
Choose four. Compare fairly.

Ten candidate climate features



Classical search

QAOA samples → SQD



Four-of-ten subset schematic · 210 possibilities

Exact classical

79.58

Uniform + SQD

80.31

QAOA + SQD

81.66

Fixed-ridge MAE · ha/fire · lower is better

Diagonal SQD returns the best sampled subset. No added optimization or quantum advantage.

Classical kernels lead.

Same four inputs. Same tuning budget.



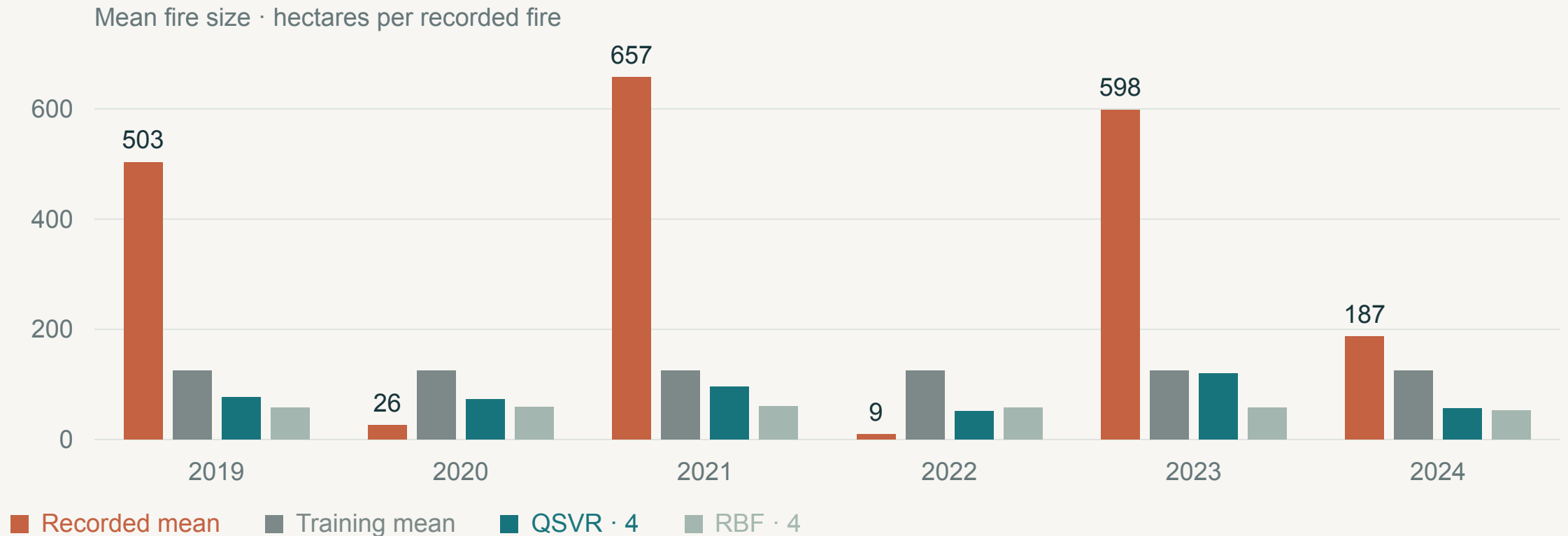
Mean chronological MAE · ha/fire · lower is better

Climate values RBF distance → SVR Qiskit state overlap → SVR

Chronological development · no significance claim

The extremes remain missed.

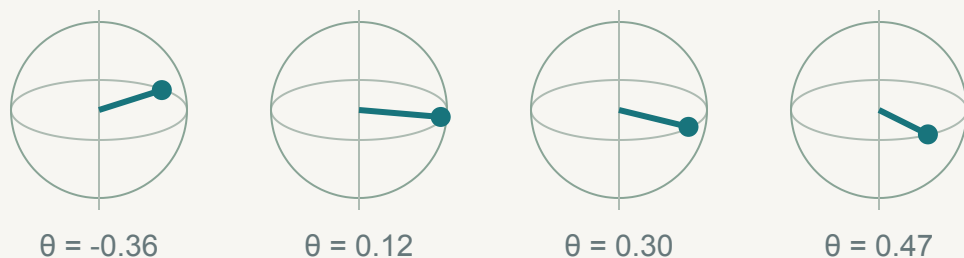
Recorded mean hectares per fire—not total burned area.



None of eleven main models beats the mean on six reused years.

Turn the angle. Change similarity.

A measured scale sweep—not a newly fitted predictor.



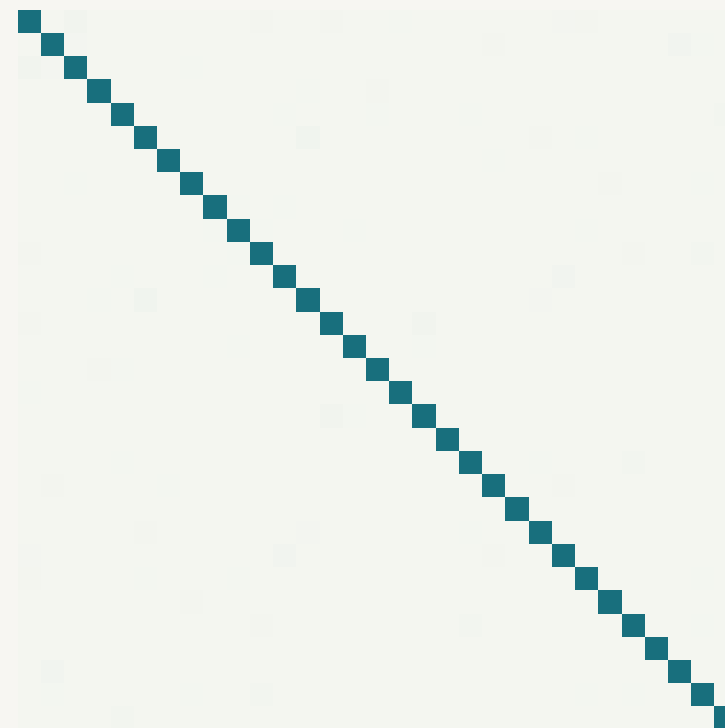
Illustrative phase rotations · radians

$\pi/32$ $\pi/16$ $\pi/8$ $\pi/4$ $\pi/2$

31.00 effective rank / 31

Mean fidelity **0.000805**

Input-only diagnostic. Narrower angles can also make kernels ill-conditioned.



Measured ten-qubit training kernel

What we can defend.

77.02 vs 86.64

RBF / QSVR development MAE · ha/fire

No later-year win.

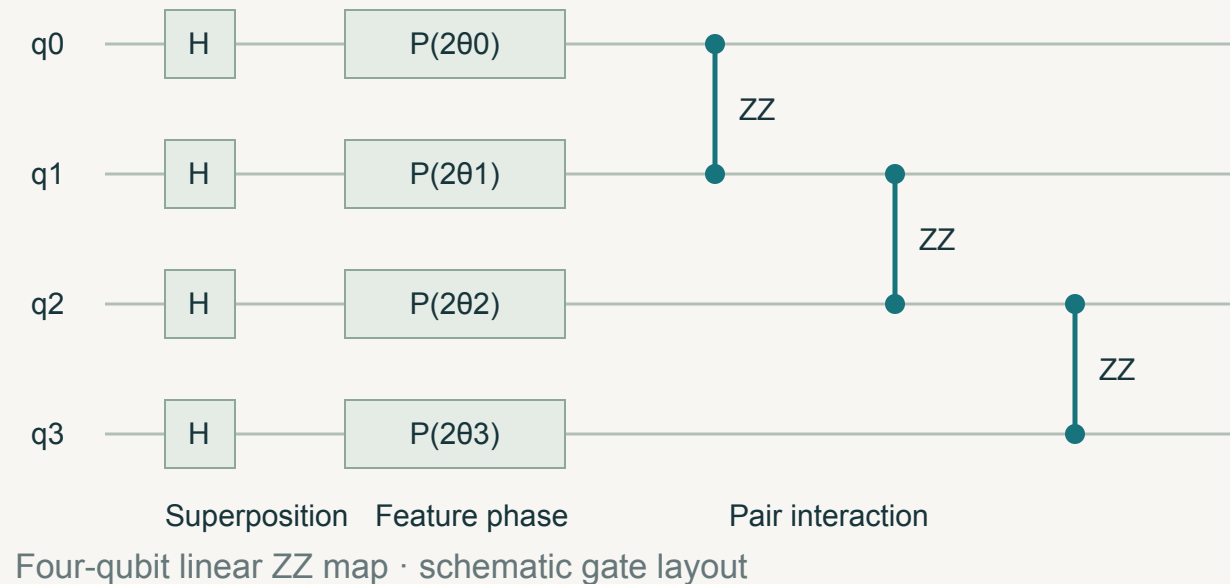
The training mean beats every main model.

Encoding matters.

Similarity changes with angle scale; skill is not guaranteed.

Next: prospective targets, independent observations and raw/log-target controls.

Measurements → phases → similarities.



Bound the standardized input

$$\theta = a \tanh(z / 2)$$

Compare the encoded states

$$k(x,y) = | \langle \varphi(x) | \varphi(y) \rangle |^2$$

Predict with a classical solver

Similarity matrix → SVR

Actual FidelityQuantumKernel + QSVR · exact simulation · training-only preprocessing
Entangling features does not guarantee predictive skill.

Small data. Explicit compute.

Final fit / saved-year evaluation · seconds

93.75 Training

9.63 Evaluation

267 kB Frozen evidence package
No-fit public replay

Fidelity pairs 12,092 train
2,232 evaluation

Scale diagnostic 4,650 pairs
48.89 s

Evaluation refits 0

Hardware jobs 0

Local analytic circuits, not physical shots. Runtime excludes installation.
Seven main slides · 300-second plan · appendices for questions.